

Abhishek Phaltankar

(413)-450-2033 | avp15102@gmail.com | LinkedIn | GitHub

EXPERIENCE

Python Developer Intern, Capgemini

Jun - Sept 2024

- Improved benchmarking accuracy by developing an ML powered vehicle comparison tool using **gradient boosting** for price prediction and **k-NN** for competitor retrieval.
- Deployed model via **FastAPI** and integrated them into a **PySide6** GUI to auto-generate PDF reports.
- Brainstormed and introduced an end-to-end workflow to streamline car and driver tracking in a testing unit.

Graduate Student NLP Researcher, UMass Amherst

Feb - May 2024

- Led the “BioLongInstruct” project, enhancing Biomedical NLP by developing long-context processing capabilities for extended medical texts using **LLMs**, including data curation for **instruction tuning** in an Agile environment.
- Implemented **LongLoRA** and other fine-tuning techniques on **Llama2** and **Gemma** models, optimizing them for complex biomedical tasks and elevating their use for extensive healthcare datasets.
- Developed benchmark tasks and tuned LLM performance on over 10 diverse biomedical datasets using prompt and instruction tuning strategies, resulting in a 9% accuracy increase.

NLP Research Intern, L3Cube

Jun 2022 - Aug 2023

- Conducted innovative research on low-resource code-mixed languages, strengthening **BERT-based language model** performance by **10%** through hyperparameter optimization with **WandB**.
- Coordinated the development of a comprehensive dataset, accurately annotating **150k sentences** for hate speech and profanity, enhancing model accuracy in identifying such content.
- Integrated language augmentation and hybrid tagging techniques developed enhanced model architectures, achieving **10-20%** improvement in classification performance.

NLP Intern, iMocha

Oct 2022 - Feb 2023

- Engineered an ML-driven Interview Assessment System using Speech to Text conversion via **Assembly AI** and training **200-500** interview question samples from various books and online resources on **Azure’s LUIS** model.
- Implemented a scoring system interpreting **5-8 intents** across 20 question types for precise assessment.
- Migrated whole project to newer **CLU framework** in response to planned deprecation of LUIS.

EDUCATION

University of Massachusetts Amherst

Master of Science in Computer Science

Amherst, MA

Sept 2023 - May 2025

University of Pune

Bachelor of Engineering in Computer Engineering

Pune, India

Jul 2019 - May 2023

PUBLICATIONS

- Challenges in Adapting Multilingual LLM’s to Low-Resource Languages using LoRA PEFT Tuning: (CHI PSAL 2025)
- Language augmentation approach for code-mixed text classification: (2023 NLP Journal by Elsevier)
- Comparative Study of Pre-Trained BERT Models for Code-Mixed Hindi-English Data: (2023 IEEE 8th I2CT)

SKILLS

Languages: Python, R, C/C++, Java, Dart, SQL, HTML, CSS, JavaScript.

Frameworks/ Tools: TensorFlow, Keras, Flask, UNITY, REST API, LoRA, Windsurf, NLTK, Git, Heroku, Power BI, Tableau, RDBMS (SQLite), Microsoft Azure, AWS SageMaker, Apache Spark, Assembly AI, RAG, MySQL, MongoDB, MLOps, OOP.

Libraries: PyTorch, Scikit-learn, Transformers, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, SciPy, SpaCy, NLTK, Hugging Face, GPT, BERT, WandB, vector DB, A/B Testing, Unit testing.